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Visualizing Flight Delay Propagation

Introduction

Currently, flight delay information is presented through airline reports, airport statistics, convoluted formulas, or simple dashboards. These show summary numbers like average delay time or cancellation rates by airline or airport. While useful, these views do not show how delays interact across the entire flight network. For example, they rarely illustrate how a delay at one airport can affect flights at other airports. Existing research studies these patterns mathematically but lack interactive visualizations that allow users to explore the data themselves.

Problem Definition

We aim to create an interactive system that helps our users understand patterns in flight delays and cancellations. By analyzing 26 million US flights, we will identify how delays vary across factors like airlines, airports, and time of day. This data will also help explore airport regions across the US (which airports serve as hubs or spokes). Our project will present these patterns through interactive visualizations so users can easily identify reliable flights, the shape of the US flight network, and how delays propagate within the US.

Literature Survey

Li and Carvalho et al. collate existing literature on flight delay propagation, summarizing machine learning, network analysis, and big-data approaches that describe how delays spread through an airline's network (Carvalho et al.; Li and Yao). These papers are useful for our project because they establish the theoretical foundation primer for analyzing delay propagation using network relationships between airports. However, the paper focuses primarily on cursory overviews of modeling and prediction rather than visualization or exploratory analysis, which is an area that we will work on. There are many different modeling methods people employ to identify how delays propagate between airports and their various causes. Dou et al. utilizes elastic net regularization to quantify the differences in delays between airlines to determine the amount "luck" or "ability" play into flight times and delays (Dou et al.). Jia et al., proposes constructing delay propagation networks using causality analysis (Jia et al.). Zeng et al. employs a cause-and-effect approach from the lens of ATC decisions (Zeng et al.). Zhang et al. propose a predictive model that incorporates spatial (i.e. distance) and temporal features to forecast airport delays (Zeng et al.). This research is highly relevant to our project because it shows there can be many ways to predict or model delay propagation. On the other hand, within the papers, there was also no one single approach towards visualizing delays, showcasing the need for a better consensus visualization tool to show results. We seek to represent this problem with a graphical approach based off the work of Wang et al. analyzing airport networks from a graphical angle (Wang et al.), corroborating through data that airlines employ a hub-and-spoke approach, similar to how Ajayi et al. uses network centrality in machine learning models to improve the accuracy of flight delay predictions (Ajayi et al.). This research is useful because it means that there exist regions of smaller airports "belonging" to a major airport. Chauhan et al., builds off this by identifying how small issues at a hub could propagate across the entire network in just 4 hops (Chauhan et al.), along with Niu et al., who introduces a network theory as a control strategy to reduce airplane delays based on node (airport) importance (Niu et al.). However, the papers primarily

focus on predictive modeling rather than understanding the structure of delay propagation, which we can improve through clear and concise visualizations. Sugishita et al., and Du et al., analyze delay propagation across Japan and China's domestic air networks using causality networks and temporal analysis (Du et al.; Sugishita et al.). Their work shows how delays tend to follow predictable patterns across connected airports. This paper is useful because it demonstrates that delay propagation can be studied through temporal relationships between airports, supporting our plan to analyze delays across time windows. However, the study focuses primarily on pattern detection rather than interactive data visualization, and any visualizations contained were difficult to parse.

Proposed Method

Our system innovates by combining a novel delay propagation algorithm with an interactive visualization platform that lets users explore the U.S. flight network in ways that existing tools do not support.

Data Pipeline: The foundation of our system is a four-stage data pipeline that processes 26,344,655 raw flight records spanning 2022 to 2025. The first stage cleans and standardizes the raw CSVs, filtering out invalid routes and deriving useful features like departure hour, producing a compressed dataset of 25,895,401 flights. Because loading 26 million rows into memory is impractical, all aggregation in the second stage runs entirely within DuckDB, a high-performance analytical query engine. This produces five precomputed summary tables covering per-airport statistics (377 airports), per-route statistics (7,565 origin-destination pairs), hourly delay distributions, per-airline aggregates, and monthly trends. The third and fourth stages build the network graph and compute propagation relationships from these summaries.

Delay Propagation Algorithm: The central novel challenge of this project is determining when a delay at one airport meaningfully caused a delay at another from open source data. Prior work such as Du et al. and Jia et al. addressed this using either private airline data or statistical causality methods that require long, dense time series. Neither is available to us: BTS (Bureau of Transportation Statistics) public data does not reliably include tail numbers, and our dataset, while large, is sparse at the individual route level. Our novel contribution is a temporal correlation algorithm that recovers propagation relationships without any of this information.

If an aircraft lands late at a hub airport and a connecting flight on the same airline departs late from that same hub shortly afterward, this is a plausible propagation event. We address this by flagging a propagation link from origin A through hub H to destination B when, on the same date and airline, a flight from A to H departs at least 15 minutes late and a subsequent flight from H to B also departs at least 15 minutes late, with the two departures separated by 30 to 90 minutes. The 30-minute lower bound prevents a flight from matching itself while the 90-minute upper bound reflects a realistic turnaround window. Running this logic as a DuckDB SQL query over the full dataset produces 9,662,750 raw propagation events, which are then aggregated into 6,695 unique hub-route summaries which capture propagation frequency, average delays, typical turnaround times, and the number of airlines affected. We then validated our results by seeing that ATL and ORD ranked among the highest propagation hubs, consistent with real-world expectations for the two busiest connecting airports in the country.

Network Graph Construction and Centrality: To understand the structure of the U.S. flight network, we build a directed weighted graph with 377 airport nodes and 7,565 edges, where each edge represents a distinct operated route that is weighted by total flight volume. We then compute five metrics for each

airport: degree centrality, weighted in-degree and out-degree, betweenness centrality, and PageRank. These metrics let us quantitatively rank airports by their influence on the broader network. We see not just how busy they are, but how central they are to connecting other airports. For community detection we use the Louvain algorithm with log-transformed edge weights, which prevents the largest hubs from dominating the result and allows the algorithm to recover meaningful regional clusters. Degree centrality represents how many airports a given airport is directly connected to by flights. In-degree and out-degree represent the number of flights arriving and departing a specific airport respectively. ATL, for example, achieves a degree centrality of 0.86 and a weighted in-degree exceeding 1.2 million flights, confirming its position at the center of the network. This is consistent with the real world, where we know that ATL is the world's busiest airport. Betweenness centrality represents how often an airport lies on the shortest paths between other airport pairs, which in the US flight network represents hubs which serve as connections between different regions or communities. DEN (Denver International Airport), has a high betweenness score of 0.164, which tracks with its real-world role as a bridge between the east and west coasts of the US. PageRank is another metric which in this context measures the importance of an airport on if it is connected to other important airports. DFW (Dallas/Forth Worth International Airport) is an example of an airport with a high PageRank score of 0.0412, an extremely high value for a graph with 377 nodes, meaning that it connects to many other major national hubs.

Visualization System: With all data precomputed, the frontend displays results instantly. The FastAPI backend loads summary tables into memory once and filters them for each user request. This helps keep response times fast even for more complex queries. The frontend is organized into four views, each designed around a task a real user might want to perform.

The Network Map places all 377 airports on a U.S. map, with nodes colored by average departure delay and sized by flight volume. Users can interactively identify whether morning or evening banks of flights at a given hub tend to run worse by adjusting sliders for the range of departure times in a day. This will then update our map with the data that the user has specifically requested. The Airport Detail view gives a deeper look at any individual airport: its centrality metrics, an hourly delay chart showing not just the average but the spread of delays across the day, and ranked lists of which airports it most frequently propagates delays to and receives propagation from.

The Delay Propagation view is the core of the project. Interactively selecting any major hub airport renders an animated radial tree showing how delays cascade outward through the network. The view consists of the root airport at the center, directly affected airports in the first ring, their downstream neighbors in the second, and so on out to four hops. We chose a radial tree because hop depth is the key variable in our propagation analysis, and having concentric rings makes that hierarchy simple for users to interpret. The tree animates one hop at a time to reinforce the sequential nature of how delays spread. Users can click any downstream airport to re-root the tree there and explore that airport's own propagation footprint, adjust a threshold to filter out weak propagation links, and hover over any node to see average delays, turnaround times, and airlines involved. This re-rooting interaction represents one of our innovations and also enables the user to to follow a delay chain wherever it leads, something no static visualization can accomplish.

The Airline Comparison view rounds out the system, letting users explore how different carriers compare on reliability and propagation behavior.

Innovations Summary: Our two primary contributions are: (1) a temporal correlation algorithm that infers hub-mediated delay propagation entirely from public flight records, without aircraft-level routing data, producing 9.6 million propagation events across 6,695 hub-route pairs; and (2) an interactive radial propagation tree that allows users to trace how delays cascade through the network from any airport, with re-rooting, threshold filtering, and hop-by-hop animation – a capability absent from existing flight delay research and tools.

Evaluation
User Qualtrics Survey

Our evaluation is designed to answer three questions: (1) Can users successfully navigate the tool and complete meaningful analytical tasks without prior training? (2) Do the visualizations communicate delay patterns and propagation relationships in an interpretable way? (3) Does the algorithm's output align with real-world expectations about which airports and airlines are most prone to delay propagation?
User Study. We conducted a user study with 6 participants drawn from peers in our program. Most regularly flew at least once a year and all were at least somewhat comfortable reading data visualizations, making them a good sample of potential users. Participants used the application directly on our machine, removing any setup burden and ensuring all participants interacted with an identical version of the tool. Each participant was given four tasks to complete: identifying the highest-influence propagation hub in the network, tracing a delay cascade through the propagation tree for a hub of their choice, using the hour filter to find peak delay times at ATL, and comparing airlines by on-time performance. After completing the tasks, participants filled out a structured form covering task completion, ease of navigation, interpretability of the visualizations, and open-ended feedback.

We asked participants if they were able to complete the four tasks and their answers were restricted to three options: yes, no, and partially. For the purposes of data analysis, "yes" answers were given a score of 2, "partially" answers were given a score of 1, and "no" were given a score of 0. The average score for identifying a major propagation delay hub was 2, indicating that participants were easily able to find these delay hubs from our U.S. Network Map visualization. Participants also had an average score of 2 for the task of tracing delay cascades through the Delay Propagation view. Users were able to find peak delay hours at a lower rate with an average score of 1.5. This indicates that users less versed in reading data visualizations might find this task difficult, as it involved moving two different slidable bars and parsing the changes in ATL's dot. For the final task, users were also able to compare airlines, with an average score of 1.333. This being the lowest score was also reflected in our gathered open-ended responses; we observed that there existed a common theme of there being too many irrelevant airlines when a majority of people were only interested in a select few airlines. However, our lowest average score being higher than a 1 indicates that users are still able to accomplish these tasks in a way that at the very least partially satisfies their needs. This finding is further reinforced by the users' responses being either somewhat or strongly agree, the two positive answer choices provided, to the following statements: "the visualizations were easy to understand", "I could navigate between views without confusion", "the information presented felt meaningful and useful", and "I would use a tool like this when planning travel or

researching flights." From our observations of the participants as they were using our visualizations, many were able to intuitively find the necessary views for each of the tasks from the left side bar. This does not mean that there were no areas for improvement though. As previously noted, we identified that people mostly struggled interpreting the Airline Comparison view because there was too much data. It was difficult to pick out the major airlines from the provided visualizations due to the sheer number of other airlines. One also felt overwhelmed by the multitude of lines crisscrossing the U.S. on the Network Map, and felt as if it detracted from the overall view. Another was also confused by our placement of the departure hour sliders on the left view when it only affected the Network Map and none of the other views. These all represented areas of improvement that we noted from our user study. However, due to the time constraints of the semester and our other classes and obligations, we were unfortunately unable to act on any of these before the deadline.

Algorithm Validation. In addition to the user study, we validated our propagation algorithm by checking whether its outputs align with well-established real-world patterns in U.S. aviation. Our algorithm ranks ATL and ORD as the top two delay propagation hubs by event count, consistent with their status as the two highest-traffic connecting hubs in the country. EWR, which is widely recognized as one of the most delay-prone airports in the U.S. due to its congested airspace and high traffic volume relative to its runway capacity, also ranks highly in our propagation output. These results give us confidence that the temporal correlation approach is capturing genuine network-level delay relationships rather than noise. We also examined propagation patterns by time of day. Peak propagation activity concentrates in the late afternoon and evening hours, which aligns with the well-known "delay accumulation" phenomenon in aviation where small morning delays compound across a day of operations. Airports serving as major connecting hubs show broader propagation footprints and affect more destinations per event compared to airports that generate fewer connecting flights.

Conclusions and Discussion

This project set out to make flight delay propagation in the U.S. network visible and publically explorable in a way that existing tools and research have not achieved. We built a complete system: from a four-stage data pipeline processing 26 million flights, to a novel delay correlation algorithm that infers propagation relationships from public data alone, to a multi-view interactive visualization that lets any user trace how a delay at one airport ripples outward through the entire US flight network.

The core finding of our algorithm validation is that meaningful propagation structure can be recovered without proprietary aircraft routing data. The 6,695 hub-route propagation relationships our method identifies reflect patterns that match our real-world expectations. Our model is able to independently calculate findings that indicate that the busiest connecting hubs generate the most propagation events, delays accumulate across the day, and certain airlines show systematically wider propagation footprints than others, purely from surface level public flight data records. This suggests that the difficulty of delay propagation research can be meaningfully lowered using the approach we have developed.

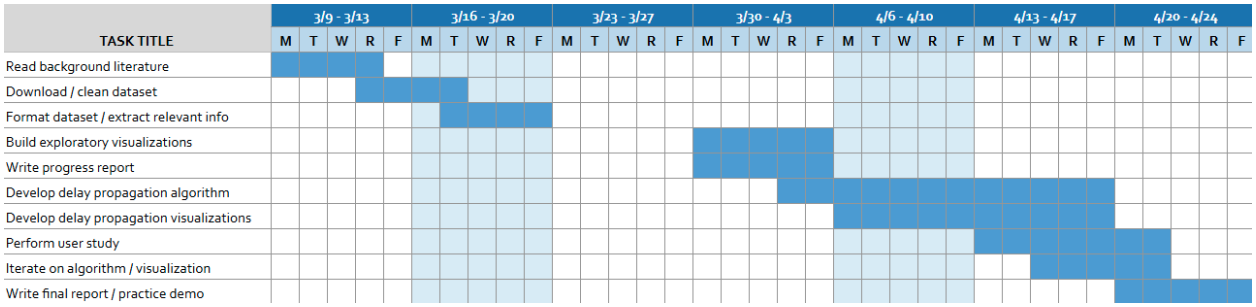
The primary limitation of our method is its reliance on hour-level time granularity rather than exact departure minutes, which means two flights on the same airline within our time window will always be flagged as a potential propagation event even if the actual aircraft involved were unrelated. Due to limitations with our dataset that did not include flight number or aircraft number information, we can only

infer relationships. This introduces false positives that a tail-number-based approach would avoid. Future work could address this by incorporating additional filtering. For example, requiring that the inbound and outbound routes share a plausible aircraft type, or weighting propagation links by the consistency of the pattern across multiple dates rather than treating each event equally.

Other natural extensions include expanding the dataset to international routes, incorporating weather event data to separate weather delays from operational ones. Another area of exploration could be to build a predictive layer on top of the current descriptive system that could forecast likely propagation given a known delay at a specific airport, time, and route. The visualization infrastructure we have built would support this extension directly. Throughout this project, all team members have contributed a similar amount of effort.

Gantt Chart

Plan:



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References

Ajayi, Joseph, et al. “Enhancing Flight Delay Predictions Using Network Centrality Measures.” Information, vol. 15, no. 9, 10 Sept. 2024, p. 559, https://doi.org/10.3390/info15090559. Accessed 31 Oct. 2024.

Carvalho, Leonardo, et al. “On the Relevance of Data Science for Flight Delay Research: A Systematic Review.” Transport Reviews, vol. 41, no. 4, 24 Dec. 2020, pp. 1–30, https://doi.org/10.1080/01441647.2020.1861123.

Chauhan, Vinod Kumar, et al. “Network Science Approach for Identifying Disruptive Elements of an Airline.” Data Science and Management, vol. 6, no. 2, 1 June 2023, pp. 110–121, www.sciencedirect.com/science/article/pii/S266676492300019X, https://doi.org/10.1016/j.dsm.2023.04.001. Accessed 28 Nov. 2023.

Dou, L, et al. “Quantifying Delay Propagation in Airline Networks,.” School of Economics Research Collection, Singapore Management University, Sept. 2024.

Du, Wen-Bo, et al. “Delay Causality Network in Air Transport Systems.” Transportation Research Part E: Logistics and Transportation Review, vol. 118, Oct. 2018, pp. 466–476, https://doi.org/10.1016/j.tre.2018.08.014.

Jia, Ziyu, et al. “Delay Propagation Network in Air Transport Systems Based on Refined Nonlinear Granger Causality.” Transportmetrica B: Transport Dynamics, vol. 10, no. 1, 20 Jan. 2022, pp. 586–598, https://doi.org/10.1080/21680566.2021.2024102. Accessed 16 Oct. 2022.

Li, N., and H. G. Yao. “A Review of Research on Flight Delay Propagation: Current Situation and Prospect.” Journal of Advanced Transportation, vol. 2025, no. 1, Jan. 2025, https://doi.org/10.1155/atr/4851103.

Niu, Xiang, et al. “From Data to Complex Network Control of Airline Flight Delays.” Scientific Reports, vol. 11, no. 1, 21 Sept. 2021, https://doi.org/10.1038/s41598-021-98112-7.

Sugishita, Kashin, et al. “Delay Propagation Patterns in Japan’s Domestic Air Transport Network.” Transportation Research Interdisciplinary Perspectives, vol. 27, Sept. 2024, p. 101235, https://doi.org/10.1016/j.trip.2024.101235. Accessed 14 Oct. 2024.

Wang, Yanjun, et al. “A Comparative Study on Flight Delay Networks of the USA and China.” Journal of Advanced Transportation, vol. 2020, 2 June 2020, pp. 1–11, https://doi.org/10.1155/2020/1369591. Accessed 9 Nov. 2020.

Zeng, Leya, et al. “Research on Delay Propagation Mechanism of Air Traffic Control System Based on Causal Inference.” Transportation Research Part C: Emerging Technologies, vol. 138, May 2022, p. 103622, https://doi.org/10.1016/j.trc.2022.103622. Accessed 26 Aug. 2022.

Zhang, Hao, et al. “A Multi‐Step Airport Delay Prediction Model Based on Spatial‐Temporal Correlation and Auxiliary Features.” IET Intelligent Transport Systems, 8 May 2021, https://doi.org/10.1049/itr2.12071.